OVER A LOCALLY NOETHERIAN BASE PSEUDO-COHERENT MORPHISMS ARE EXACTLY THE LOCALLY FINITE TYPE ONES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17472-17477 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17479-17490, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted A side only (spaces twin proof incomplete per ruling v3).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemma Noetherian-pseudo-coherent.

Source reference. Original lemma 066E: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17490–17490. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. If S is locally Noetherian, then f is pseudo-coherent if and only if f is locally of finite type.*

Proof. This translates into the following algebra result: If $R \to A$ is a finite type ring map with R Noetherian, then $R \to A$ is pseudo-coherent if and only if $R \to A$ is of finite type. To see this, note that a pseudo-coherent ring map is of finite type by definition. Conversely, if $R \to A$ is of finite type, then we can write $A = R[x_1, \ldots, x_n]/I$ and it follows from More on Algebra, Lemma 066E that A is pseudo-coherent as an $R[x_1, \ldots, x_n]$-module, i.e., $R \to A$ is a pseudo-coherent ring map. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.